

Frequently Asked Questions for FQ_xI Foundational Review Series and Foundational Answers

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1. FAQs

1. Q: What is the Foundational Review Series (FRS)?

- FRS is a nontraditional journal for high-impact reviews of topics in foundational physics, broadly defined as research that addresses the most fundamental questions about the universe’s governing laws on the nature of space, time, and matter. The scope of FRS therefore encompasses, but is not limited to, the following fields: quantum gravity, quantum foundations and information, cosmology (namely inflation, dark matter, dark energy, cosmological constant, fine-tuning, and multiverses), the nature of time, agency, and the role of generative AI in reshaping foundational physics.

FRS aims to create central reference texts for foundational physics by providing works that are self-contained, and serve as repositories of the current state-of-the-art and accessible introductions for FQ_xI members, postdoctoral researchers, and graduate students in theoretical physics, interested in learning about a foundational topical area. Moreover, we envision these manuscripts as a space for authors to offer informed physics-based speculations from their own perspective.

2. Q: What is the difference between FRS and FQ_xI’s Foundational Answers Program?

- While manuscripts from the Foundational Answers Program will be published within the FRS platform and both initiatives are dedicated to synthesizing knowledge within foundational physics, they operate with distinct structures, scopes, and objectives within FQ_xI.

FRS operates as a journal dedicated to high-impact living reviews across a broad spectrum of foundational physics. FRS manuscripts serve a dual purpose: acting as

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comprehensive repositories of the current state-of-the-art for the advanced non-expert (e.g., theoretical physics graduate students) interested in learning about foundational topics, while explicitly providing a venue for authors to offer informed, physics-based speculations from their own theoretical perspectives.

By contrast, The Foundational Answers Program is a targeted programmatic initiative. Rather than functioning as an ongoing journal, it actively engages specific teams of academic researchers to produce synthetic reviews on topics that are centrally aligned with FQ_xI priorities. Here, the primary goal is operational consolidation by creating a repository of the knowledge gained from specific research efforts so that it may serve directly as a springboard for future collaborative work.

3. Q: What is the AI policy of FRS?

- Authors are expected to acknowledge and cite the usage of Artificial Intelligence (AI) large language model (LLM) model used at any point in the manuscript preparation. It is sole and exclusive responsibility of the authors for the integrity, accuracy, and originality of their entire manuscript. The use of AI or AI-assisted technologies does not diminish this responsibility. Any errors, factual inaccuracies, or ethical breaches are the on the authors, regardless of whether they were generated by AI. That being said, FRS does not prohibit the use of AI-assisted editing tools for grammar, spelling, or formatting purposes

4. Q: Does FRS accept unsolicited proposals?

- While we appreciate hearing about interesting ideas at the cutting-edge of foundational physics, submissions and manuscript proposals to FRS are currently by invite only.

5. Q: Is there an arXiv pre-print policy?

- Authors may simultaneously submit their manuscripts to both ArXiv and FRS. Following formal FRS review and editorial processing, authors are encouraged to update their arXiv pre-prints to reflect the peer-reviewed text and scientific corrections to make the pre-print as scientifically accurate as possible. While authors are expected to update their text to match the final accepted version, specific visual illustrations commissioned by FQ_xI, such as professional illustrations or custom informative graphics, may not be uploaded to arXiv. Authors must use their own original figures and diagrams in the arXiv pre-print version. Upon formal publication on the FRS platform, author must update their arXiv metadata to include the official citation and a direct link to the manuscript's open-access landing page on the FQ_xI FRS Platform.

6. Q: What is the target audience and expected technical level of an FRS manuscript?

- The target audience for an FRS manuscript is broad, ranging from theoretical physics Ph.D. student to tenured faculty and other FQxI members. Therefore, the technical level should be at that of a theoretical physics Ph.D. student who is interested in learning about a foundational physics topic. The review should consist of a conceptual “on-ramp” to the topic to get the motivated reader to a level of understanding such that they can understand the current state-of-art in the field and the points made by the author.

7. Q: Since the manuscripts in FRS by invite only, how is a topic and author chosen?

- Topics are decided based on the perceived need for a review on foundational topics for our intended audience (see previous question) or if we aware of a eminent and renowned scientist(s) working on a new approach to a problem. At present, the vast majority of invited authors are FQxI members, but this need not be the case. Since FRS is a new journal, FQxI is establishing the journal’s credibility by boasting its impressive membership. That being said, if a chosen review topic is not represented well by FQxI’s membership, FRS has and will invite non-member scientists who are recognized experts in the field of topic.

8. Q: Does the review process differ from the traditional approach?

- Actually, yes it does! In these earliest stages, the review process will primarily go about the usual way for referees. However, during the review, we ask referees to flag statements or calculations that they believe should be verified. In these cases, an independent scientist, such as a graduate student or postdoctoral fellow, will be contracted to reproduce the result and note any differences arising between their approach and the authors. These independent scientists will also be given substantial (non-author) credit for their work. Finally, we also kindly ask referees for any suggestions on possible independent verifiers for the claims.
- FRS also operates under a transparent peer-review model where all formal referee reports may be published with the living review and referees will be given notable non-author credit for their contribution to improving the manuscript. As a reviewer, you will have the choice to choose whether or not you would like your name to be published within the document and the review.